

Geomec 5.7 Release Notes

Geomec 5.7.1 has the following improvements:

- 1) Bug fixes, among others in SKUA import, Temporary path popup at start, Rockmech
- 2) Additional improvements for Rockmech error message handling

Geomec 5.7.0 has the following new features and improvements:

- 1) Mesh refinement for CM2 tetrahedron mesher
 - Create rotatable, scalable refinement boxes with adjustable gradation
- 2) Double-sided faults (each side with different discretization) has been extended:
 - Creates an intermediate surface (optionally); results are mapped back to the input surfaces
 - Uses stress mapping for improved calculation results
- 3) Improvements to well and casing zoom-in models
 - New selectable boundary stiffness based on parent model's Young's and Shear modulus
 - Visualization of parent input values (stress, pressure, displacement, boundary stiffness)
- 4) Improved import of Earth meshes (SKUA format)
 - Import formation (zone) and volume (compartment) names
- 5) Improved error message handling for Rockmech
 - Detect if a model contains input for depletion stages higher than initial
 - Mark more informational messages as fatal
- 6) Resolved stress initialization problems in mixture mode calculation
- 7) Allow multiple installations of Geomec to co-exist
- 8) Speed, stability and other improvements
 - Faster loading of a model
 - Faster generation of the well zoom-in mesh
 - Resolved crashes during:
 - i. switching between parent and child models
 - ii. well export
 - iii. switching between well and history view
 - iv. deleting a well zoom-in model that has a previously opened casing model
 - v. scaling/deformation use in casing model
 - vi. changing cement material in-between calculations
 - Improved some dialogs for readability
 - Added originating point and element set names to distributed property attributes
 - Resolved bug for order of values in the wellpath view
 - Resolved bug where remeshing didn't preserve volume assignment to formations
 - Export of well results in main model and clipboard copy of wellpath view are now more similar

Known issues:

- 1) Mesh refinement does not include surface re-meshing
- 2) Stress/strain calculation artifacts exist in well and casing models with surfaces intersecting semi-parallel to axial direction

Known issues from 5.6 still present in 5.7:

- 1) The Python Plugin system doesn't work with Shell's Enthought Python distribution version 7.2.2
- 2) Only one plugin can be run during a Geomec session

Known issues from 5.4 still present in 5.7:

- 1) GVT is an alpha test version. It doesn't support export/import of velocity files.
- 2) Qt 5.6.2 has problems with handling symbol characters. As a workaround, "°C" is currently rendered as "C", but there may be more issues.
- 3) The option to show convex hulls doesn't include cylindrical hulls yet.

Known issues from 5.3 still present in 5.7:

- 1) Contour lines are absent in the legend, and can only be shown for the property as the shading shows
- 2) Using graph ranges in the 2D wellpath view may clip the wellpath on the right, depending on Geomec's window size